

Visual Search Advantage for Illusory Faces

A Computational Reproduction Study

Background: The Face Detection Advantage

The human visual system has evolved an extraordinary sensitivity to **faces**.

Key Observations

- Faces capture attention in visual clutter automatically
- We can detect a face faster than we can find an equally visible non-face object
- This happens across diverse face images (photos, drawings, schemas)

The Paradox: Face Pareidolia

- Humans perceive faces in **inanimate objects** (clouds, toast, electrical outlets)
- These "illusory faces" lack traditional face features (skin texture, facial structure)
- Yet they still trigger rapid face-detection mechanisms

The Research Question

Do illusory faces confer visual search speed advantage as real human faces?

Why This Matters

1. Tests the specificity of face-detection mechanisms
2. Reveals whether face perception depends on low-level features or high-level configuration
3. Informs theories of evolved attentional biases and evolutionary trade-offs

Hypothesis

Face pareidolia leverages the same search advantage seen with real faces, despite lacking facial features.

Experiment 1: Grid Search (Homogeneous Distractors)

Design Goal: Test illusory face detection among visually similar objects.

Setup

- **Visual Layout:** Invisible 8×8 grid (64 positions max)
- **Set Sizes:** 16, 32, or 64 items
- **Stimulus Size:** 120 × 120 pixels (each image)
- **Distractors:** All from the *same object category* as the target (e.g., if target is a cheese grater, all distractors are unique cheese graters)

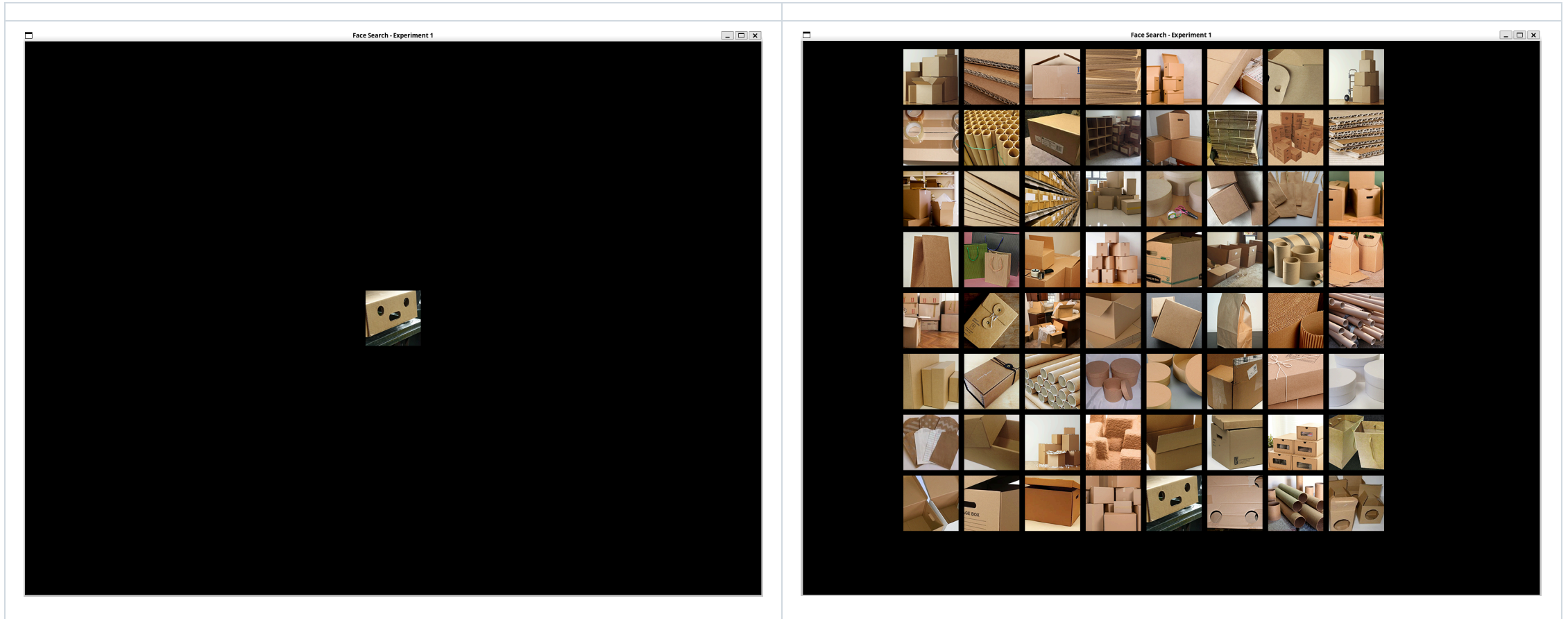
Factors

Factor	Levels	Design
Target Type	2	pFace vs. nonFace
Target Presence	2	Present or Absent
Set Size	3	16, 32, or 64 distractors
Categories	26	One per participant (repeated 12×)

Trial Count

- 26 categories × 12 trial types = **312 trials**
- Organized into **6 blocks** (breaks every ~52 trials)

Example: Ex1



Experiment 2: Circular Search (Heterogeneous Distractors)

Design Goal: Lower task difficulty and directly compare real faces vs. illusory faces.

Setup

- **Visual Layout:** Equidistant circular array
- **Set Sizes:** 4, 8, or 16 items
- **Stimulus Size:** 120 × 120 pixels (with circular mask)
- **Distractors:** Objects from *diverse categories*, not matching the target

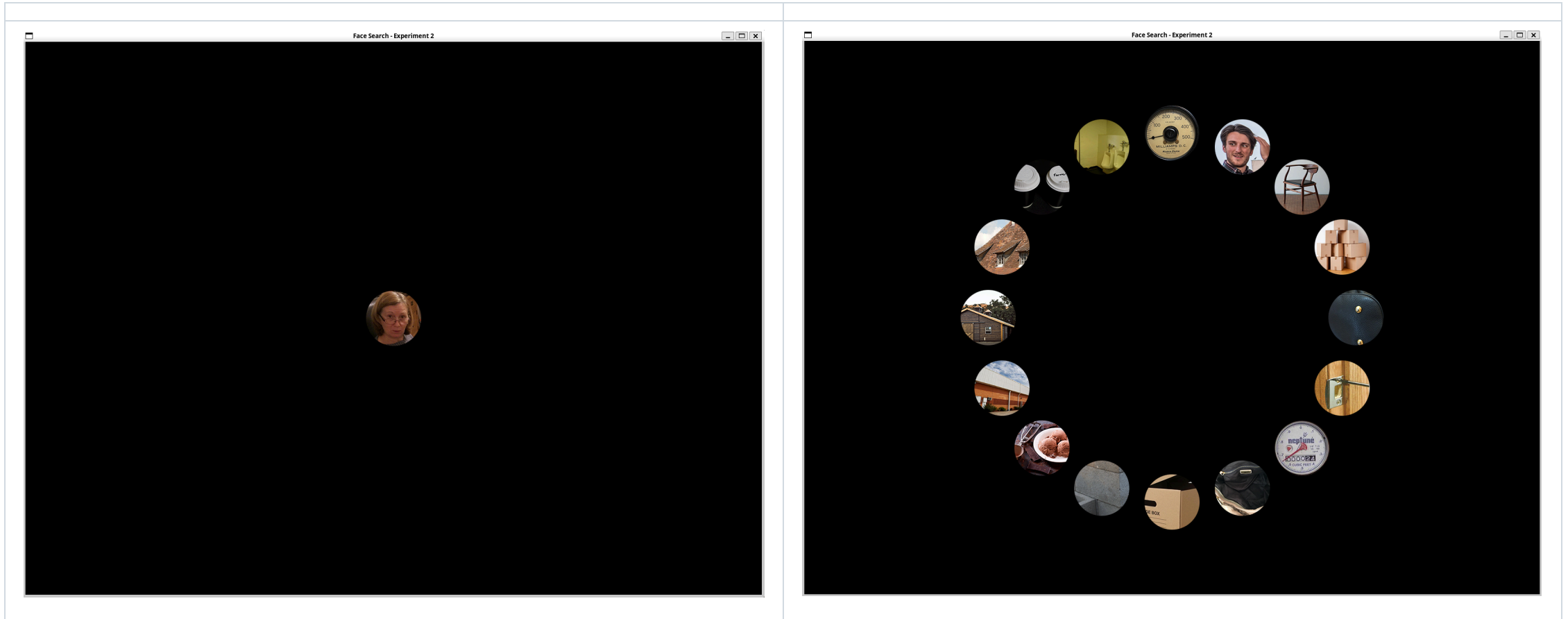
Factors

Factor	Levels	Design
Target Type	3	nonFace, pFace, or realFace (human face)
Target Presence	2	Present or Absent
Set Size	3	4, 8, or 16 distractors
Categories	23	One per participant (repeated 18×)

Trial Count

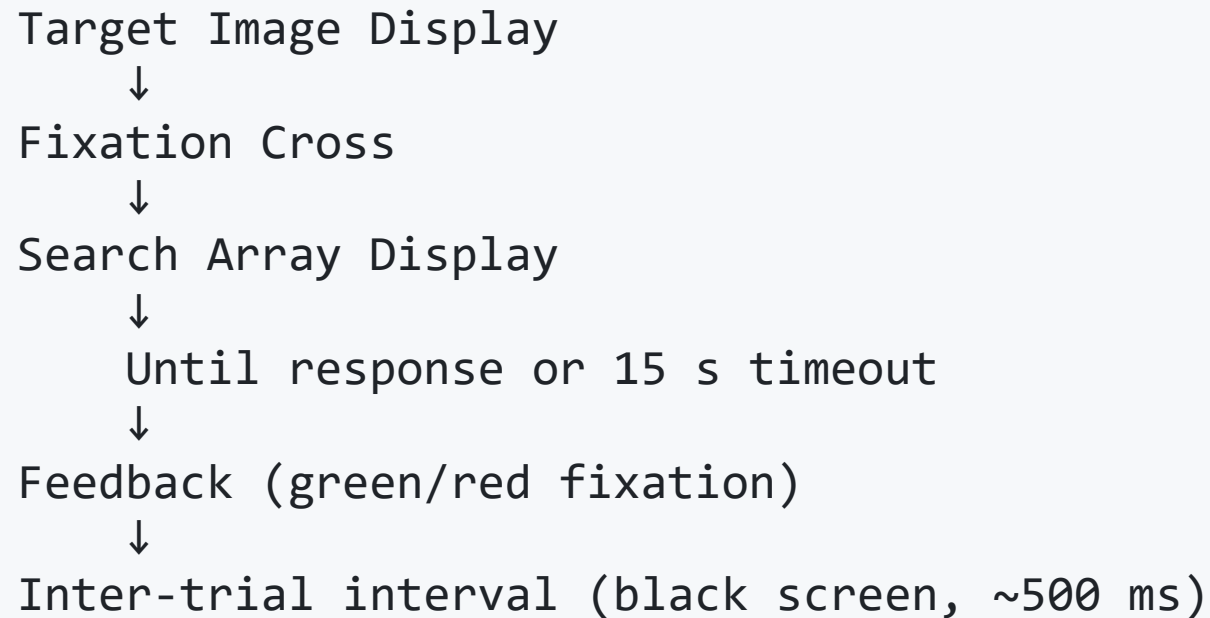
- 23 categories × 18 trial types = **414 trials**
- Organized into **6 blocks** (breaks every ~69 trials)

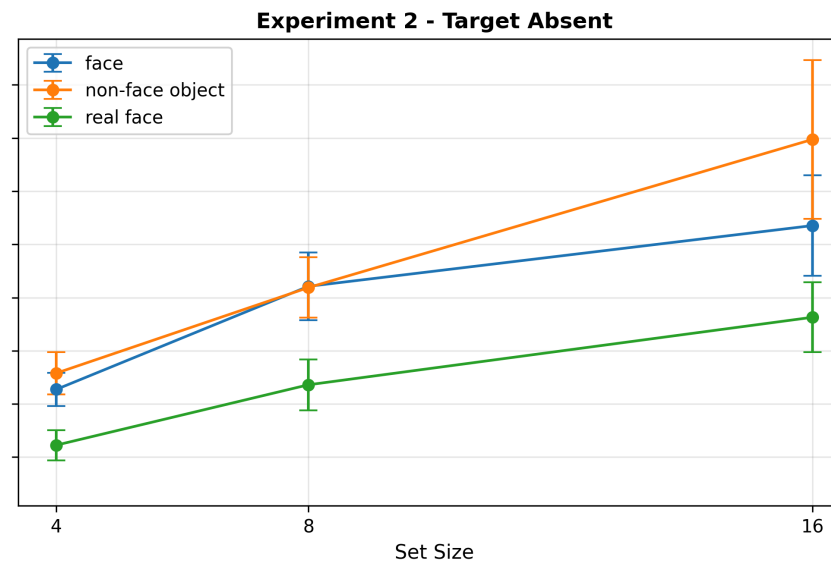
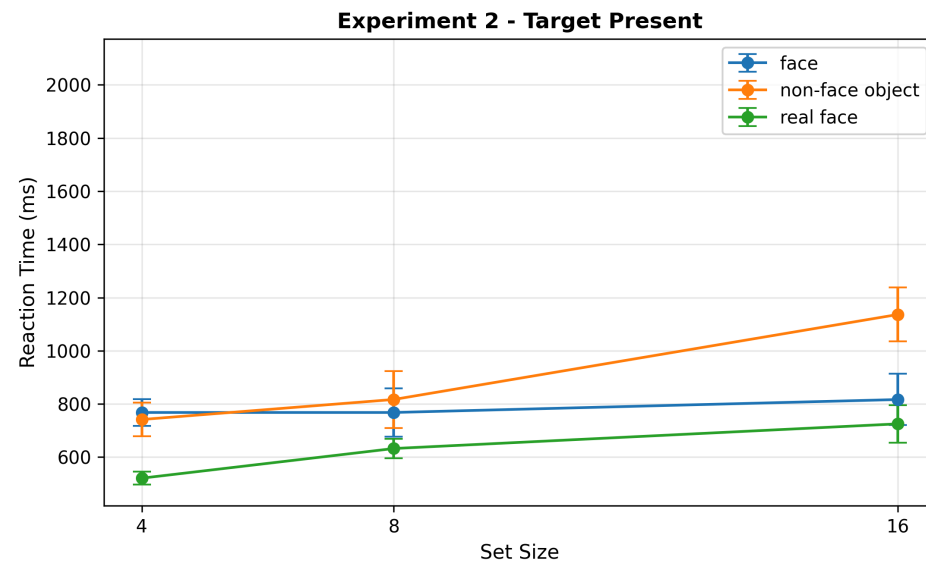
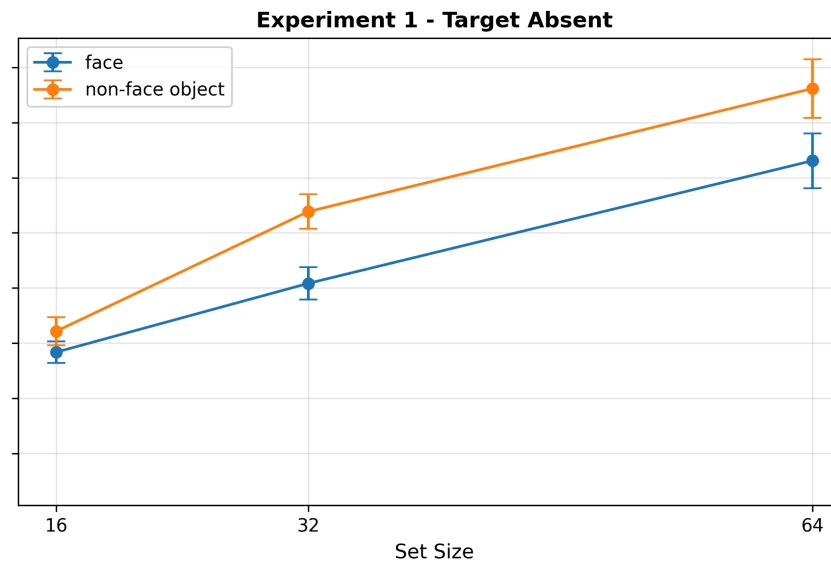
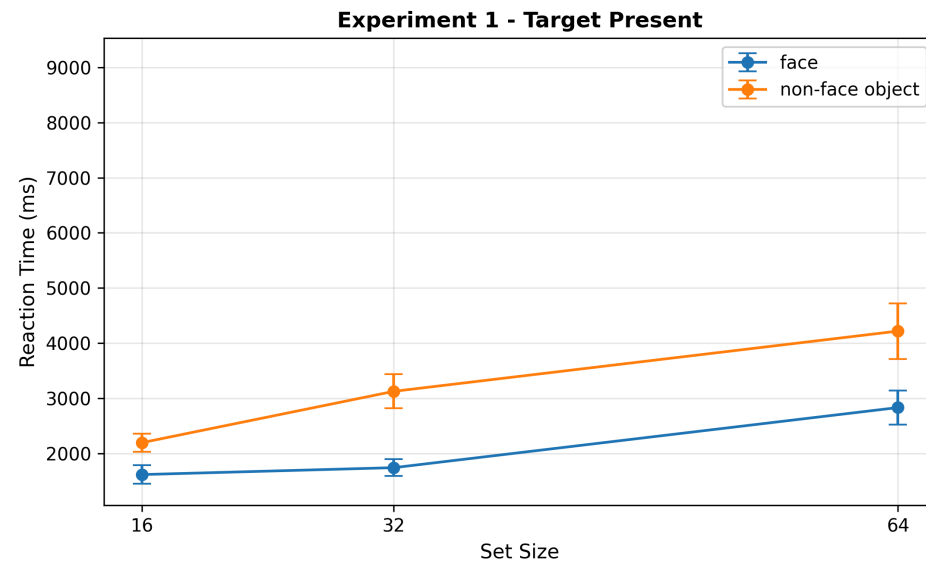
Example: Ex2

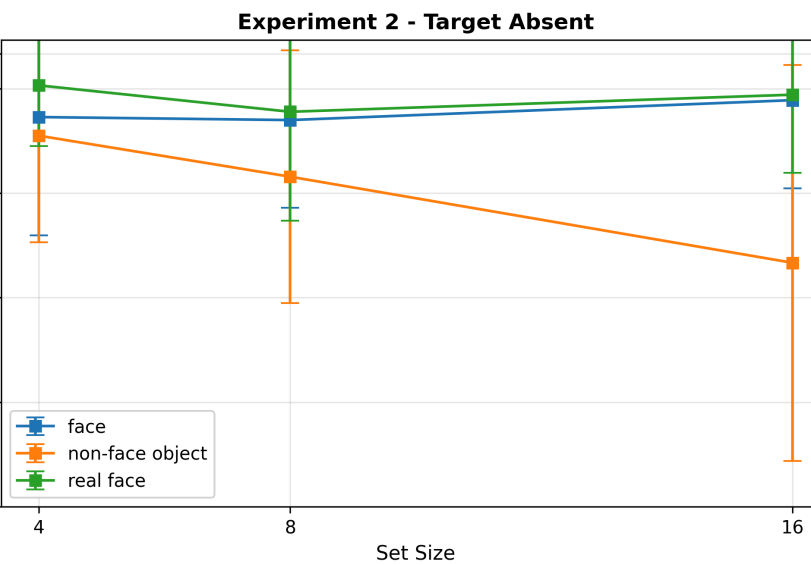
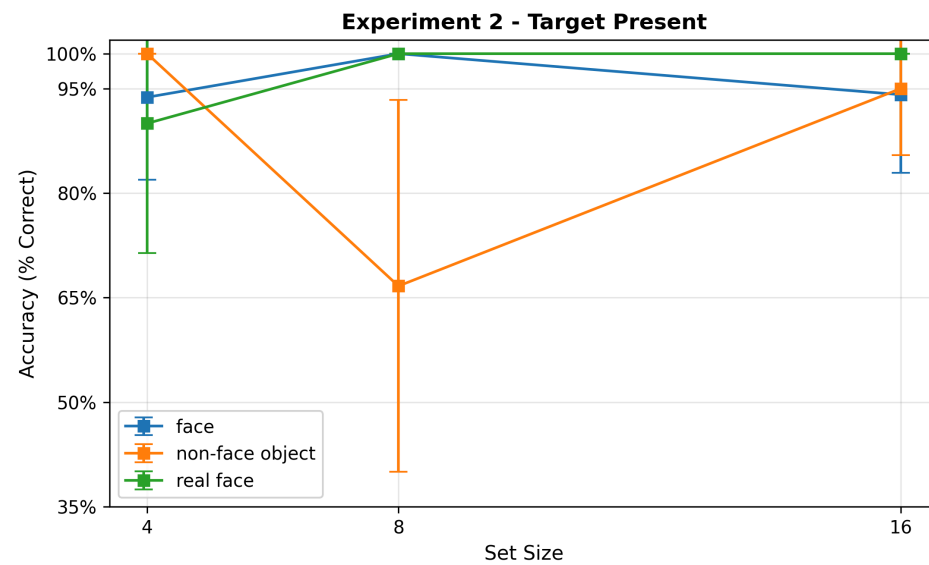
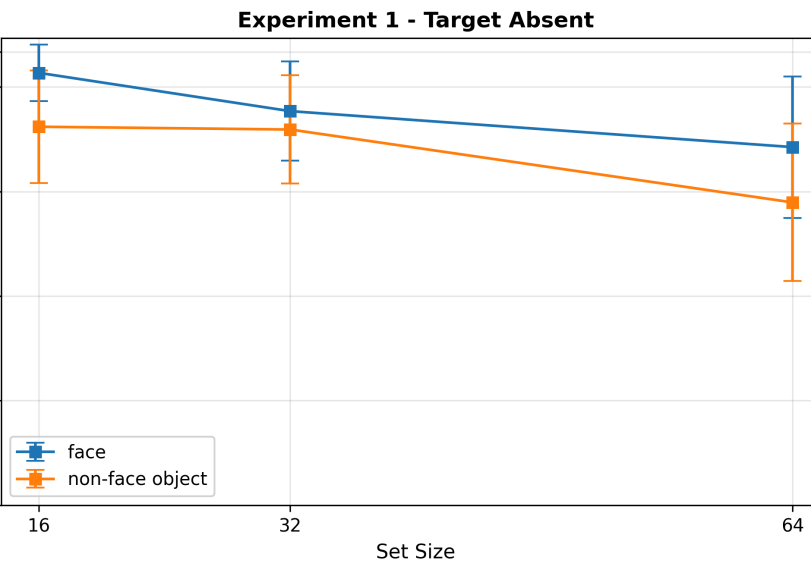
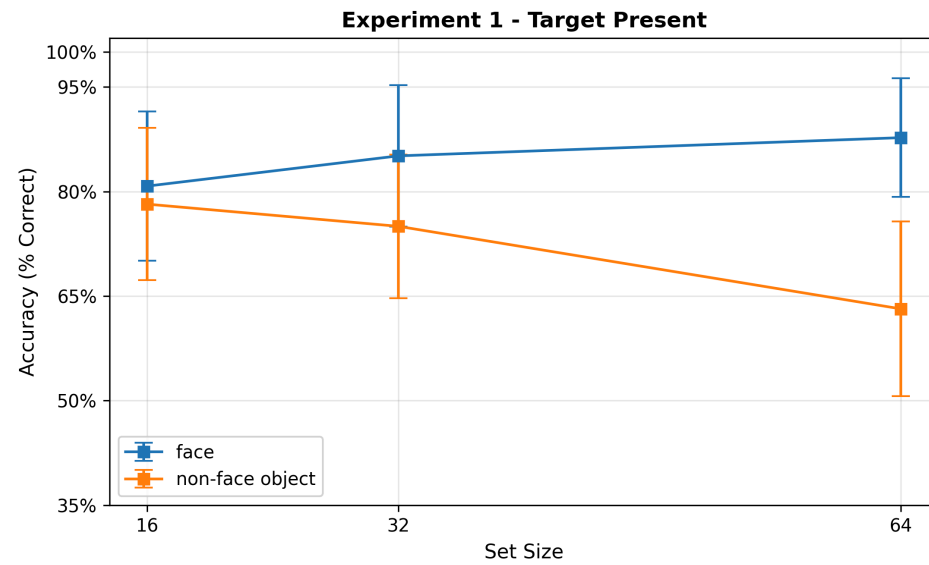


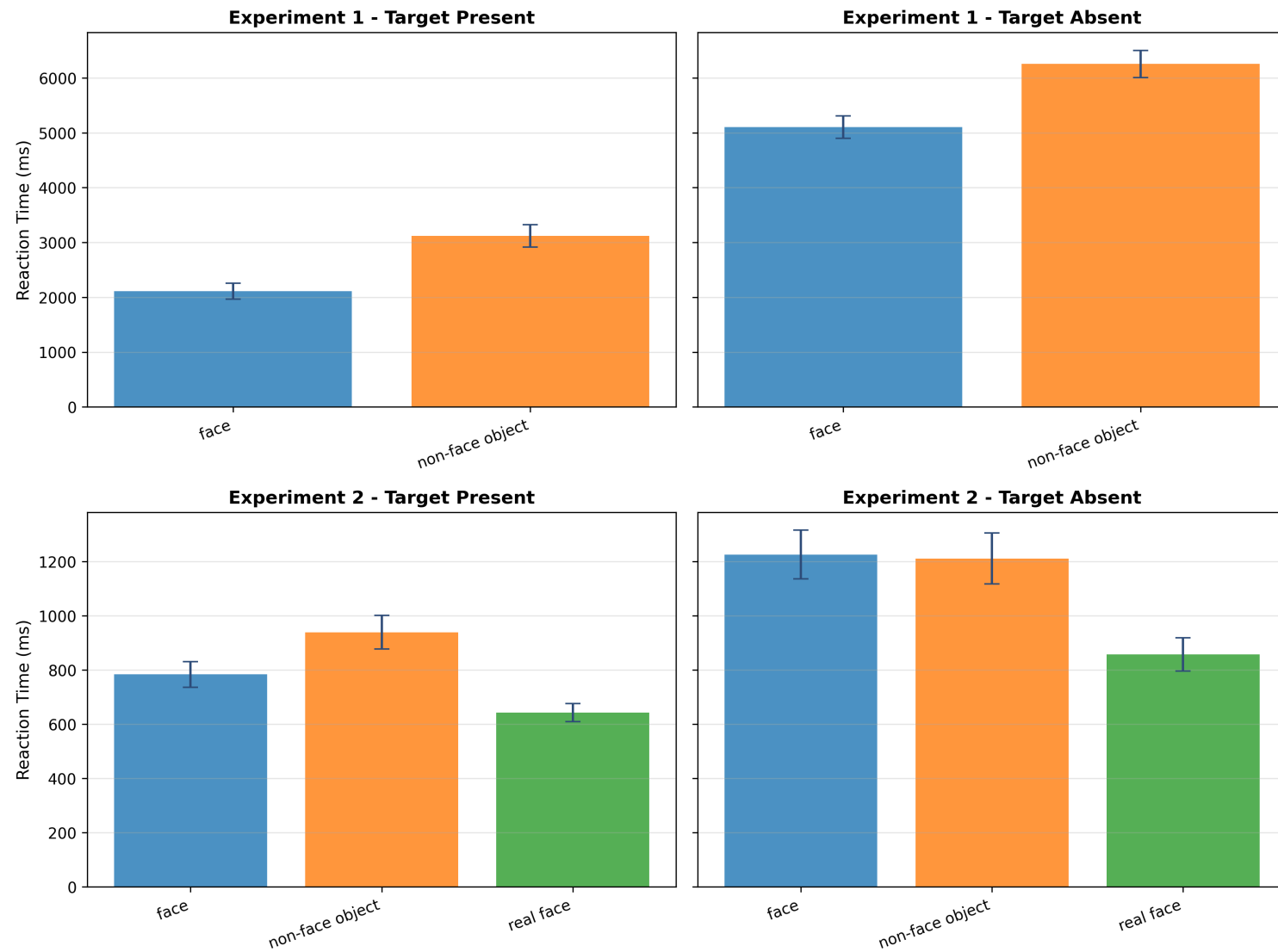
Timing & Trial Structure

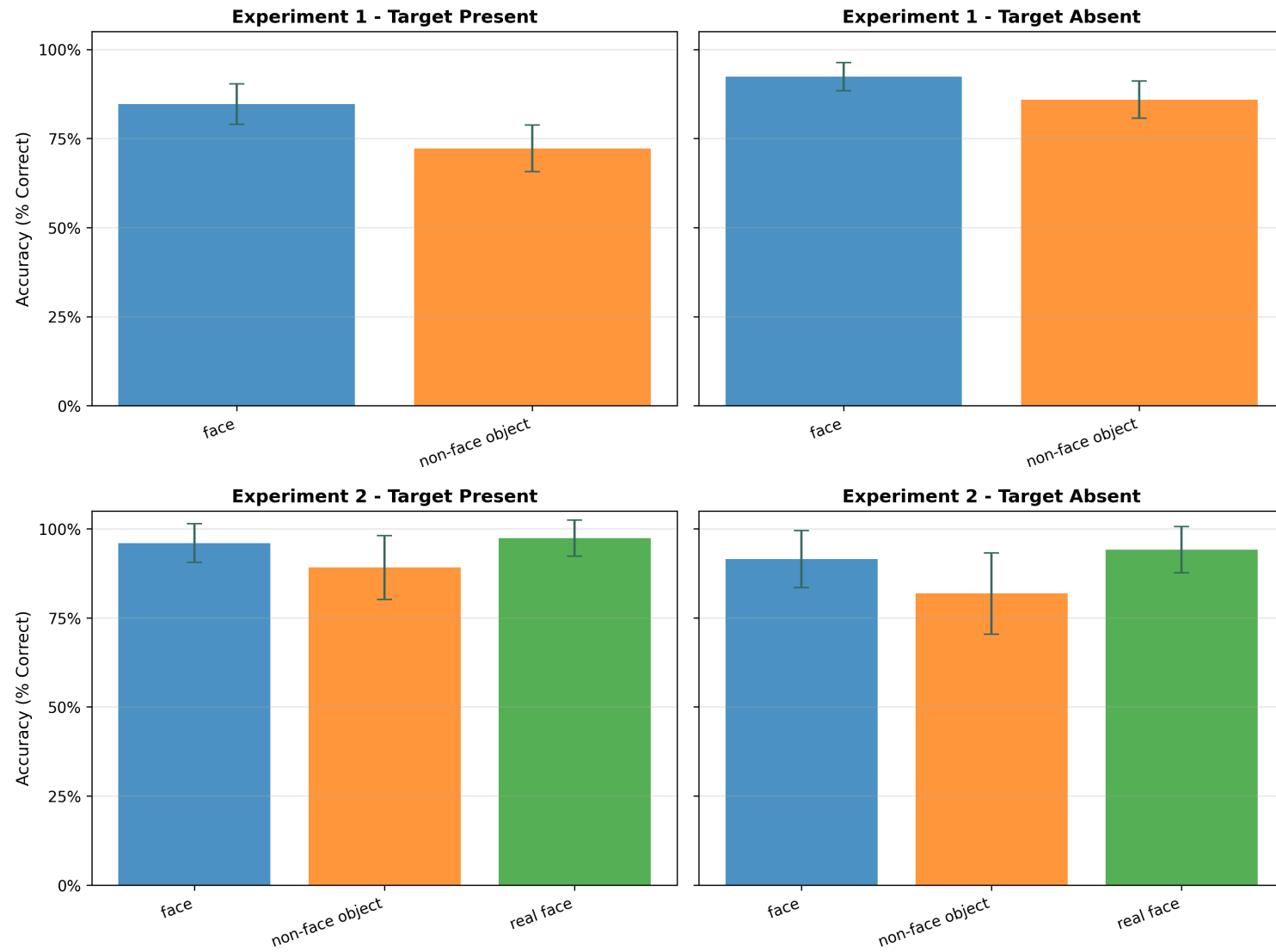
Each trial follows a **precise temporal sequence**:

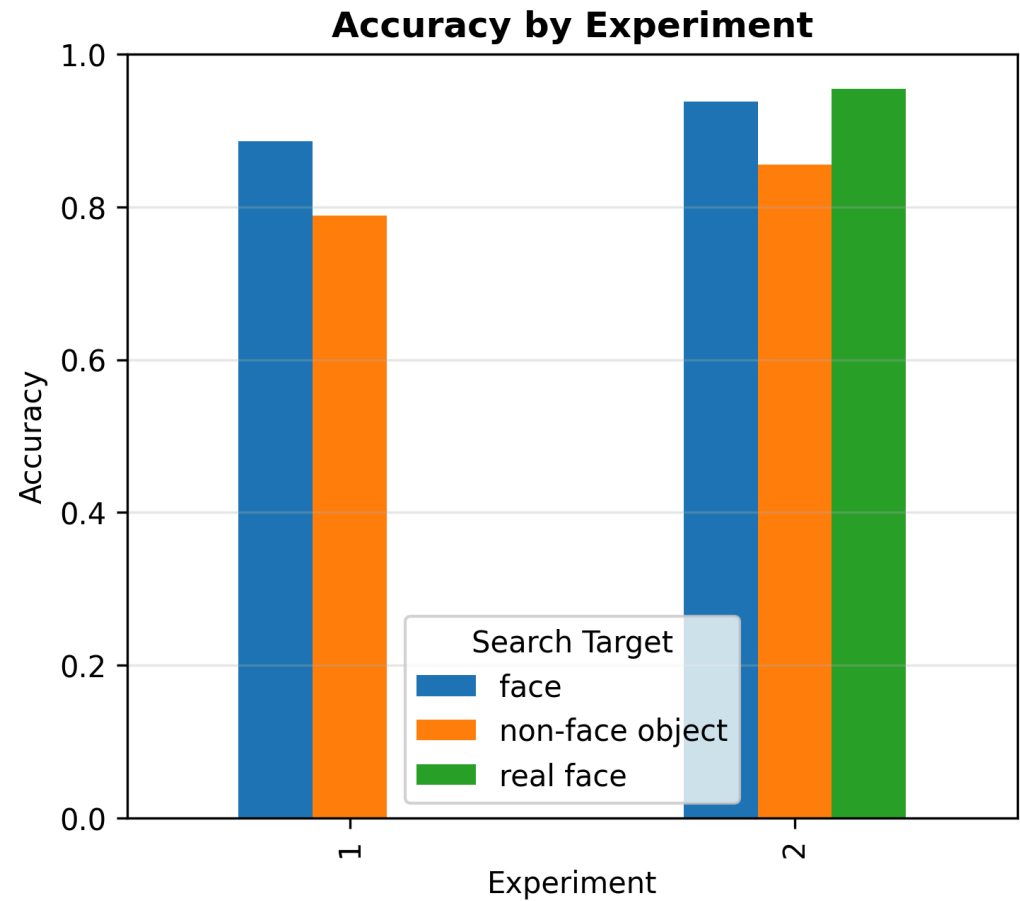
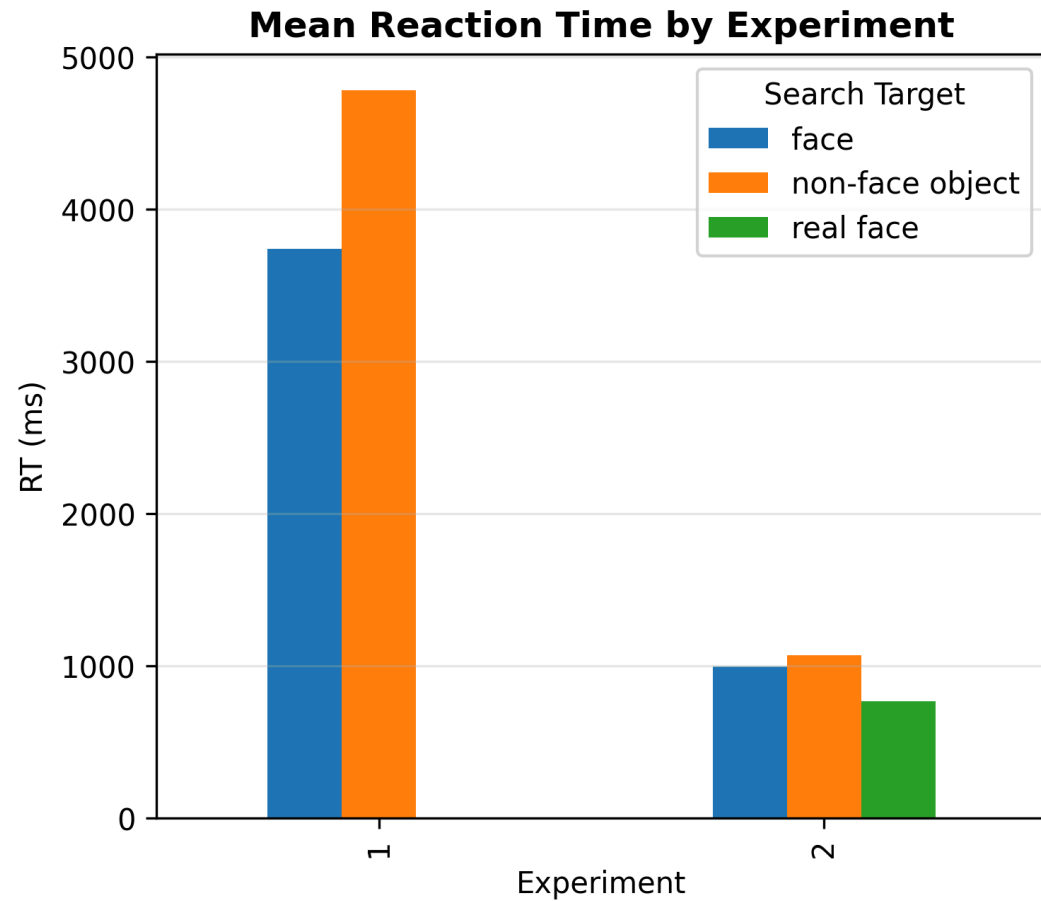












Observed Findings

Experiment 1

- Main effect of target type: illusory faces faster than non-faces, $F(1,17)=32.09$, $p<.0001$ (Target-Present)
- Main effect of set size: larger sets yield slower responses, $F(2,34)=28.58$, $p<.0001$
- Significant interaction in Target-Present: $F(2,34)=8.85$, $p=.0008$
- Target-absent trials overall slower than target-present trials

Observed Findings (Cont.)

Experiment 2

- Main effect of target type: Real Faces < Illusory Faces < Non-Faces, $F(2,14)=15.56$, $p<.001$ (Target-Present)
- Main effect of set size: smaller sets yield faster responses, $F(2,14)=15.07$, $p<.001$
- Significant interaction in Target-Present: $F(4,28)=9.27$, $p<.0001$
- Target-absent trials overall slower than target-present trials

Observed Findings (Cont.)

Theoretical Implication

Search advantage for illusory faces suggests a **broadly-tuned, configural face detector**:

- Does not require low-level face features (texture, color)
- Responds to any spatial configuration resembling a "face-like" layout
- Implies evolutionary advantage of detecting any face-like configuration over missing a real face

Implementation: Python + Pygame

Why Python?

-  Cross-platform compatibility (Windows, macOS, Linux)
-  Open-source (no proprietary software dependency)
-  Precise timing via `pygame` for response collection
-  Reproducible data collection with random seed control

Key Technical Features

1. Stimulus Yocking (Ex1):

- All distractors sourced from same category as target
- Ensures homogeneous visual context

2. Circular Layout (Ex2):

- Equidistant positions computed trigonometrically
- Stimulus mask applied to enforce circular presentation

Key Technical Features (Cont.)

3. Deterministic Randomization:

- Category order avoids adjacent repeats
- Trial type order randomized within each category
- Reproducible via random seed storage

4. Data Recording:

- Response times to millisecond precision
- Trial order saved for full reproducibility
- JSON manifest + CSV output for analysis

Data Analysis Pipeline

Raw CSV Output



Correctness Filtering (remove timeout errors)



Reaction Time Analysis

- ANOVA: Target Type × Set Size × Target Presence
- Planned contrasts: pFace vs. nonFace; realFace vs. pFace vs. nonFace
- Visual search slopes (RT vs. Set Size)



Accuracy Analysis

- Conditional error rates by target type



Visualization

- Line plots: RT vs. Set Size (by target type)
- Bar plots: RT and accuracy comparisons

Conclusion: Implications for Face Perception

What This Tells Us

1. **Configurational Face Detection:** Humans possess a face-detection system tuned to *spatial configuration*, not low-level features.
2. **Evolutionary Hypothesis:** This broad tuning reflects an adaptive trade-off—the cost of false positives (pareidolia) is negligible compared to missing a real face in social or threat contexts.
3. **Neural Substrates:** Illusory face detection likely engages similar neural populations (fusiform face area, amygdala) as real face detection.

Thank You!

Questions?

This reproduction demonstrates that illusory face detection is indeed a genuine visual search advantage, supporting decades of face perception research through computational replication.